The Impacts of Teacher Turnover: A Comparative Study of Public and Charter Schools in Illinois

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Abstract

Teacher turnover may detrimentally impact student outcomes by reducing overall teacher quality, disrupting the learning environment, or forcing diversion of a school's limited resources. However, given variation in operational practices between public and charter schools, effects of turnover may differ; the greater flexibility of charter schools could allow them to more effectively dismiss underperforming teachers, improving teacher quality and thereby student outcomes. Using panel data on school demographics, student performance, and teacher retention from the Illinois State Board of Education over the years 2014 to 2023, I examine the effect of teacher turnover on multiple student outcomes. I analyze heterogeneity in these effects between public and charter schools, as well as between demographic groups. I find statistically significant evidence to suggest that turnover diminishes standardized test proficiency rates in public schools yet has a null effect in charter schools. Effects are heterogeneous by race, with Black and white students being more detrimentally affected by turnover while Latino and Asian students may, in some cases, benefit. In addition, turnover decreases average SAT reading scores in public schools and increases both average SAT math scores and the high school dropout rate in charter schools. Beyond statistical significance, the magnitude of these effects is larger than those found in existing literature, suggesting findings may be economically meaningful and policy relevant.

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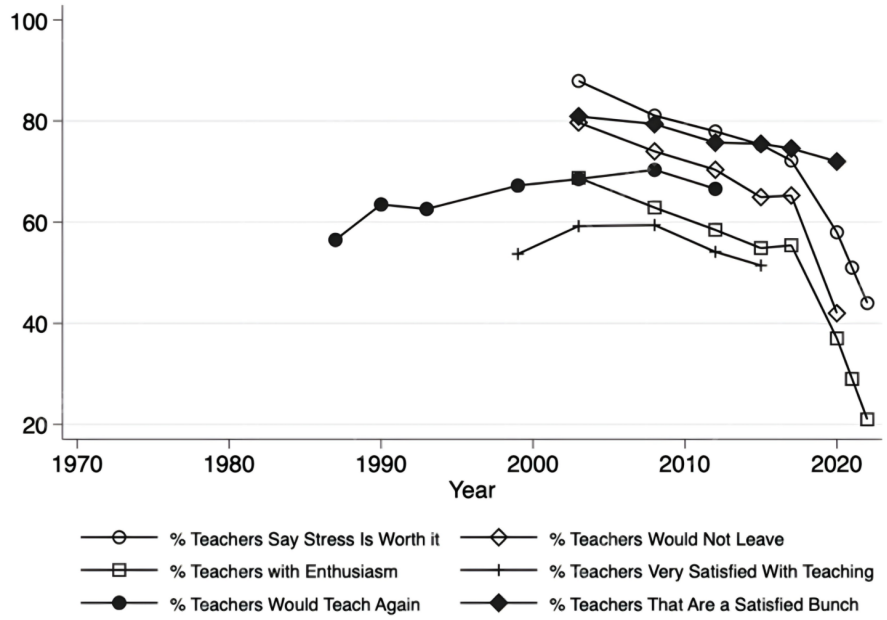
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1 Introduction

1.1 The State of the Teaching Profession

From Plato to Paulo Freire to Eve Ewing, thinkers across both time and disciplines have emphasized the formative role that educators play in human society. Yet by many accounts, the teaching profession in the United States is on the decline. As displayed in Figure 1, measures such as teachers' enthusiasm for teaching, the percentage of teachers who feel the stress of their career is worthwhile, and teachers' overall job satisfaction are all at historic lows (Kraft and Lyon, 2022). This trend may be explained in part by low compensation relative to other careers: The average salary of a U.S. public school teacher in the 2021-22 school year was $66,745, only marginally higher than the national average. While teacher pay has historically been higher than the national average due to the education, certification, and other credentials often needed to enter the profession, this gap has closed in recent years (National Education Association, 2023). Moreover, teachers report a variety of poor working

Figure 1: Measures of teacher satisfaction in U.S. public schools over time



Source: Status of the American School Teacher, National Center for Education Statistics, RAND American Teacher Panel, American Federation of Teachers Member Survey (2022)

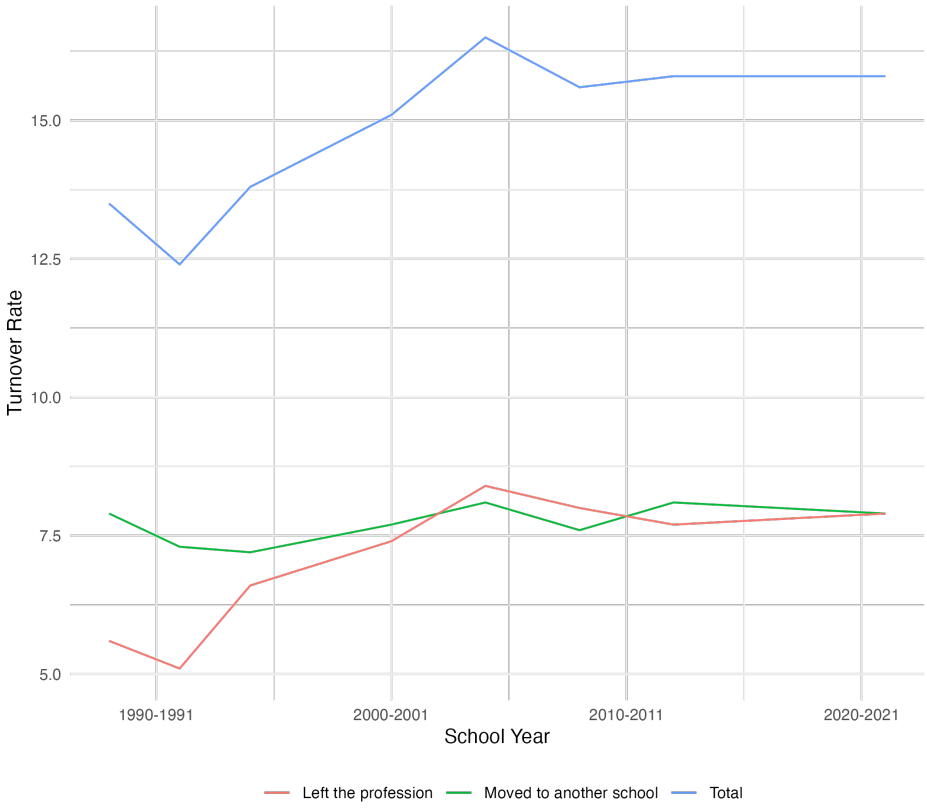
conditions, including a low degree of autonomy over their lesson plans (74.5 percent), a lack of support from school administration (50.4 percent), and threats from students (21.8 percent), according to a report from the Economic Policy Institute (García and Weiss, 2019). Recent events – such the COVID-19 pandemic and increased levels of political polarization around topics such as race and gender – have only added further workplace stressors (Woo et al., 2022).

Given this context, it is unsurprising that the U.S. currently faces a national shortage of teachers. In January of 2022, 16 percent of public schools nationwide reported a teaching vacancy in mathematics – that is, the inability to hire a full-time mathematics teacher where one was desired – and 13 percent of public schools reported a teaching vacancy in English Language Arts (Schmitt and deCourcy, 2022). In October of 2023, 86 percent of public schools reported some form of teacher staffing difficulty (Modan, 2023). Teacher turnover rates, as measured by the number of teachers voluntarily or involuntarily leaving their school at the end of the academic year, also remain consistently high. As shown in Figure 2, the average annual turnover rate of kindergarten through 12th grade public school teachers in the U.S. is approximately 16 percent, a number virtually unchanged since 2000; of this 16 percent, about half leave the profession entirely (National Center for Education Statistics, 2022). These shortages and high rates of turnover may have detrimental effects on students: The direct costs associated with turnover, such as hiring and training new teachers, may divert resources and lead to underinvestment in certain areas. Moreover, indirect costs such as the resulting changes to the composition of a school's teaching population and shifting coworker dynamics may negatively affect the performance of individual teachers.

1.2 The Rise of Charter Schools

Aside from these persisting issues surrounding teacher retention, another much-discussed topic in education policy is that of charter schools. Minnesota passed the nation's first charter school law in 1991, allowing for the creation of City Academy, the first public charter school, in 1992 (Schroeder, 2004). Then, in 1994, the Clinton administration created the Federal Charter Schools Program as an amendment to the 1965 Elementary and Secondary

Figure 2: Teacher turnover rates in U.S. public schools over time

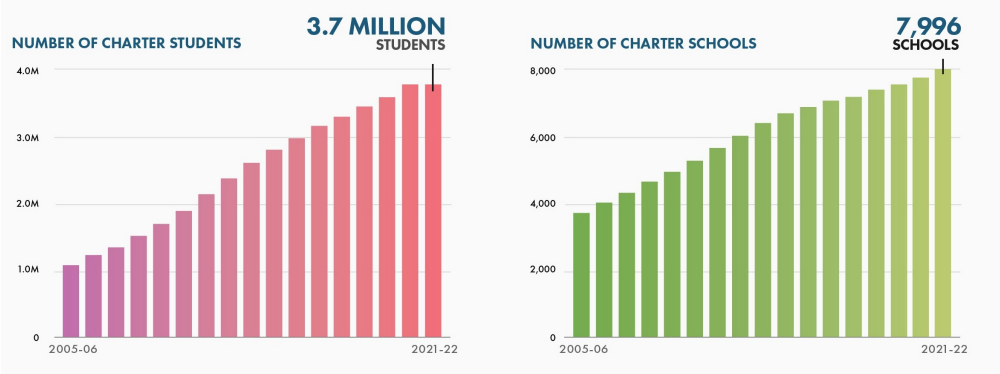


Source: National Center for Education Statistics (2022)

Education Act, designating government funding for charter schools. From there, the number of public charter schools across the nation skyrocketed, and by 2022, 3.7 million students attended nearly 8,000 public charter schools across 45 of the 50 states, as displayed in Figure 3 (White, 2023).

Public charter schools (henceforth referred to as "charter schools") are distinguished from traditional public schools (henceforth referred to as "public schools") by their higher degree of autonomy. While charter schools receive government funding, they operate independently of public school districts. The founding of a charter school begins when an individual or group forms a Charter Management Organization (CMO) and draws up an academic mission, funding plan, and other logistic details for a new school. The CMO then submits an application to a charter school authorizer; while types of authorizers vary by state, they

Figure 3: Charter students and charter schools in the U.S. over time



Source: National Alliance for Public Charter Schools (2023)

often include nonprofits, school districts, and governmental authorities. If approved, the CMO then receives public funding and begins to prepare for the school's opening by hiring faculty, creating curriculum, establishing an educational facility, and more. Throughout this process, the CMO has more flexibility in spending its funds than public schools, whose financial support may come in the form of fixed resources such as chairs, desks, and books (National Charter School Resource Center, 2019). In addition, while the CMO is held to a legislative contract called a charter that outlines certain requirements it must follow – including public enrollment, anti-discrimination laws, core subject offerings, and regular reviews by the charter school authorizer – it is also free to operate outside of many public school district guidelines. For instance, charter schools do not have set hiring dates or pay rates for teachers, may evaluate and dismiss employees according to their own set of standards, have control over their instructional materials, and are not bound to district budget allocation guidelines (Ableidinger and Hassel, 2010).

The charter school educational model has been subject to both praise and criticism from U.S. educators, policymakers, and economists since City Academy's inception. Some studies, such as those evaluating the Knowledge Is Power Program (KIPP) charter school network, have found that charter schools improve student outcomes by boosting college enrollment rates and helping to close the racial and socioeconomic achievement gaps (Angrist et al., 2011). Evidence of test score gains in Boston charter schools (Abdulkadiroğlu et al., 2011)

and a diminished racial achievement gap in Harlem charter schools (Dobbie and Fryer Jr., 2011) have provided further evidence in favor of charters.

However, others argue that the movement for school choice – that is, the push to provide alternative educational options to public schools – may harm public schools and their students by siphoning off students and resources, especially as public schools often receive funding on a per-pupil basis. From 2010 to 2022, while the number of charter schools increased by nearly 3,000, the number of public schools fell by a similar amount (National Center for Education Statistics, 2022). Critics frame such a shift as alarming, claiming that charter schools facilitate a transition into privatized education and that their greater operational flexibility prevents them from being held sufficiently accountable to producing the best student outcomes. Moreover, charter schools also face the same teacher shortage and turnover problems as public schools, often to an even greater extent. The average annual turnover rate in charter schools is approximately 24 percent, significantly higher than that of public schools (Jason, 2017). Still, this is not necessarily a negative, as differences in the quality of such turnover are unobserved; that is, turnover may have differential impacts on student outcomes between charter and public schools.

1.3 Roadmap

The paper proceeds as follows: Section 2 provides an overview of the relevant literature and identifies remaining gaps. Section 3 details the theoretical framework involved in modeling the impacts of turnover. Section 4 describes the data and methodology. Section 5 presents and analyzes the results. Section 6 concludes.

2 Literature Review

A large body of economic literature provides evidence that teacher turnover negatively impacts student outcomes, though such findings focus primarily on analyses of public schools and almost exclusively measure outcome using standardized test scores. I briefly discuss some of these key papers, as well as the current understanding of turnover in charter schools. I then outline my intended contributions to the literature.

2.1 Teacher Turnover in Public Schools

Ronfeldt et al. (2013) identify a significant and negative effect of turnover on standardized test scores by studying fourth and fifth grade public school students in New York City from 2001 to 2010. The authors apply two specifications, one using school-by-year fixed effects and one using school-by-grade fixed effects, to capture variation across grades within a given school and year. By examining the effect of grade-level turnover rates on individual-level student outcomes, they conclude that a change from 0 percent turnover to 100 percent turnover diminishes student math scores by about 0.082 standard deviations and student English Language Arts (ELA) scores by about 0.049 standard deviations. The larger impact on math scores is consistent with literature suggesting that math test scores are generally more responsive to instructional changes than ELA scores, as math learning tends to be more cumulative and is more limited to a classroom setting than literacy (Anderson, 2022). Additionally, these effects are particularly strong in schools with a higher population of low-performing and Black students. The authors find evidence to support a compositional explanation of turnover; that is, turnover harms student outcomes in part because it leads to a change in the distribution of teacher quality by replacing more effective teachers with less effective ones. When controlling for measures of teacher ability, such as years of experience and previous value-added scores – which measure the difference between a student's expected and actual performance – the remaining effects of turnover are diminished. However, the paper also suggests the compositional hypothesis does not have full explanatory power, as even with the additional controls, the effects of turnover remain significant and negative. Though these effects remain unexplained, the authors hypothesize that they may stem from the way turnover could disrupt a school's sense of community or climate, as well as the financial burden imposed on schools to hire and train new teachers, both of which may then impact teacher performance and thereby instructional quality. These mechanisms are discussed in further depth in Section 3.

Similarly, Hanushek et al. (2016) examine an urban school district in Texas over the years 1996 to 2001 and find that student scores on the Texas Assessment of Academic Skills

(TAAS) diminish as a result of teacher turnover. In keeping with Ronfeldt et al. (2013), the authors model turnover using both school-by-grade and school-by-year fixed effects. While they find turnover to be a detriment, their estimates are of a smaller magnitude, with math standardized test scores decreasing by about 0.045 standard deviations when the turnover rate changes from 0 percent to 100 percent. Notably, this effect is observed most strongly in lower-achieving schools, while there is little to no evidence of adverse turnover effects in higher-achieving schools. The authors also conduct an analysis of the effectiveness of different subgroups of teachers: those who move to another school in the same district, those who move to another school but in a different district, and those who exit the Texas public school system entirely. The latter group, as measured by value-added scores, appears to be significantly less effective than the other two, providing support for the possibility that turnover could have a positive effect via negative selection of poor-performing teachers out of schools and the profession more generally. Yet, the authors find that despite this negative selection, new hires may be even less effective; in addition, controlling for teacher experience nullifies the findings in the primary specifications, providing evidence to support Ronfeldt et al. (2013)'s compositional hypothesis.

Thirdly, Sorensen and Ladd (2020) also find turnover detrimental to student achievement, and to an even higher extent than Ronfeldt et al. (2013) and Hanushek et al. (2016). Using North Carolina public school data from the years 1994 to 2016, they find that an increase in turnover from 0 percent to 100 percent decreases the standardized test scores of middle school students (grades 6 through 8) by 0.072 standard deviations in ELA and 0.134 standard deviations in math. Moreover, they further bolster the compositional hypothesis by finding significant changes to the teaching population as a result of turnover. An increase in turnover from 0 percent to 100 percent increases a school's proportion of teachers with 0 to 3 years of experience by 38.0 percentage points, increases a school's proportion of teachers with a lateral or provisional license (non-traditional certifications that tend to be correlated with weaker teacher performance) by 15.4 percentage points, and increases a school's proportion of teachers teaching outside their subject area by 3.7 percentage points. As found in the

prior two papers, these effects are strongest among lower-performing schools and those with more economically disadvantaged students. Also in line with the prior papers, the authors observe these causal effects by applying a fixed-effects regression that models student scores in a given subject, school, and year – or school-level teacher population outcomes in a given year – as a function of turnover, student characteristics, school subject, and school and year fixed effects. However, one notable difference is that the authors measure turnover via a three-year running average, citing research such as Holme et al. (2018), who note that such a measure is likely more stable and can better capture effects driven by administrative responses to longer-term periods of instability.

The Ronfeldt et al. (2013), Hanushek et al. (2016), and Sorensen and Ladd (2020) papers all suggest that teacher turnover harms student test scores, possibly by changing the composition of teaching staff and thereby the instructional quality, with heterogeneity by school achievement level. Yet, this analysis of school type does not extend to charter schools. I seek to replicate their findings on more recent student outcomes data with a broader variety of outcome measures, as well as apply their analysis to investigate whether turnover has differing impacts in public as compared to charter schools. Moreover, Hanushek et al. (2016) introduce the possibility of negative selection as a mechanism for turnover to improve teacher quality and thus student performance, a hypothesis that ultimately does not hold up in public schools but may be more applicable under the more flexible guidelines of charters. And, as in Sorensen and Ladd (2020), I will model turnover with a three-year average to better account for effects of turnover stemming from administrative changes.

2.2 Teacher Turnover in Charter Schools

Statistically, turnover in charter schools differs significantly from that in public schools. Stuit and Smith (2012) use 2003 to 2004 data from the National Center for Education Statistics to examine both quantitative and qualitative dimensions of these differences. Applying a nonlinear version of the Oaxaca-Blinder decomposition, they find that charter schools faced a turnover rate of about 24.2 percent, significantly higher than the 11.9 percent turnover rate in public schools. Moreover, this gap was largely explained by differences in educa-

tor characteristics: Charter schools had a significantly higher proportion of inexperienced teachers (37.2 percent as compared to 16.3 percent in public schools), uncertified teachers (14.6 percent to 1.4 percent), and part-time teachers (14.0 percent to 8.6 percent), all of whom were more likely to turn over. Yet, the single most important variable was union membership, which explained 18.7 of the gap. While 82.1 percent of public school teachers were union members, this percentage was only 24.0 among charter school teachers. The authors did not directly investigate the mechanism behind this effect but did hypothesize that it may be driven by two factors. First, teachers may leave charter schools in search of more stable, unionized positions. Charter and public school teachers in the sample reported similar salary satisfaction, but teachers leaving charter schools were more likely than public school teachers to cite salary and benefits as key factors in their decision. Secondly, charter schools' greater independence from unions may allow them more operational flexibility in dismissing underperforming teachers. In accordance with this hypothesis, the study found significantly higher rates of involuntary attrition in charter schools: 14.9 percent of total turnover as compared to 5.9 percent in public schools. Still, while the study identified a variety of differences between public school and charter school turnover, it did not investigate the effects of such turnover on student outcomes.

In a policy brief examining teacher turnover in New Orleans, where the majority of schools are charters, Barrett et al. (2020) find stronger evidence of negative selection than in neighboring public schools. That is, in New Orleans, teachers ranked in the bottom 20 percent in terms of value-added measures are 2.5 more times more likely to leave their position than teachers ranked in the top 20 percent. Meanwhile, in neighboring districts, this ratio is 1.9. The results suggest that charter schools may be able to improve teacher quality by selectively retaining their high-performing teachers. Yet, the authors do not find any evidence of a changing gap in teacher quality between New Orleans schools and neighboring districts, leading them to conclude that the entrance of new teachers – who tend to be of lower quality – offsets the potential advantages of turnover. While the brief draws attention to the mechanisms behind turnover, it primarily focuses on how turnover impacts teacher

quality rather than directly addressing effects on student outcomes, which I aim to do.

On the whole, there is relatively little literature that focuses on the differential impacts of teacher turnover on student outcomes in charter schools as compared to public schools. Naslund and Ponomariov (2018) examined 2012 to 2013 data from the Texas Educational Agency on public school districts and charter school organizations to conclude that in charter school organizations, the negative effect of turnover on student achievement is reduced in comparison to public school districts. Using an ordinary least squares regression with interaction terms, the authors found that an increase in turnover rates from 0 to 100 percent diminished a public school district's student standardized test pass rate by 28.9 percent (math), 28.6 percent (science), and 31.3 percent (writing). The effect on math and science pass rates was mitigated in turnovers – a 14.3 percent decrease in math and a 9.1 percent decrease in science. The authors also investigated the impact of turnover on high school graduation rates and found a null effect, both in public schools and in charters. However, a number of areas of investigation remain. Notably, the authors did not explore the mechanism behind these effects, nor did they determine whether the dampened impacts of turnover may be canceled out by charter schools' higher turnover rates. In addition, their analysis used a single year of data with district-level observations; to avoid endogeneity concerns, I aim to replicate their findings at a more granular level and incorporating further years of data.

2.3 Contributions to the Literature

This paper's contribution to the literature is threefold. First, it extends existing research on the impacts of turnover on standardized test scores to more recent Illinois state data. It also extends this analysis to charter schools and examines heterogeneity in the effects of turnover, further fleshing out Barrett et al. (2020)'s account of changing teacher quality and building upon Naslund and Ponomariov (2018) by conducting a more granular, longitudinal investigation of student outcomes. Lastly, this paper seeks to understand how turnover affects student outcomes beyond the scope of test scores. I follow Naslund and Ponomariov (2018) in examining the effects of turnover on high school graduation rates and also introduce two new measures of achievement: student attendance rates and high school dropout rates.

3 Theoretical Background

Ronfeldt et al. (2013), Hanushek et al. (2016), and Sorensen and Ladd (2020) all find evidence in favor of a compositional explanation of the effects of turnover; that is, turnover negatively impacts student achievement in part because it decreases instructional quality by replacing more experienced teachers with less experienced, and therefore less effective, ones. While not directly linked to student outcomes, Barrett et al. (2020) support these hypothesized changes to teacher quality. Ronfeldt et al. (2013) also suggest that turnover may have disruptive effects operating at a level beyond that of the individual teacher. In particular, turnover could impose financial burdens on schools to hire and train new teachers, lead to a sub-optimal reorganization of remaining teachers, or change peer effects among teachers – i.e., the performance of one teacher may be influenced by their coworkers – all of which may diminish resulting instructional quality. Thus, to understand how teacher turnover impacts students, it is first important to construct a model of instructional quality, then extend this to a model of student outcomes.

3.1 Modeling Instructional Quality

Throughout the economics of education literature, a common measure of instructional quality is teacher value-added, briefly introduced in Section 2.1. Calculated by subtracting a student's expected performance from their actual performance, value-added estimates a teacher's impact on student standardized test scores (Lovenheim and Turner, 2018). To control for selective assignment of students to teachers, the model relies on lagged test scores that control for students' prior ability. Equation 1 details the full calculation process.

$$\Delta score_{igsty} = \alpha + \theta_t + \delta_g + \rho_s + \tau_y + \beta X_{gsty} + \mu Z_{iy} + \varepsilon_{igsty} \tag{1}$$

The change in test score for student i in grade g and school s with teacher t during year y is regressed on teacher fixed effects θ_t, grade fixed effects δ_g, school fixed effects ρ_s, year fixed effects τ_y, a vector of observed classroom characteristics X_{gsty}, and a vector of time-varying student characteristics Z_{iy}. The estimates of θ_t are the teacher value-added measures. One

key critique of using this metric to gauge quality is that student test scores may not be the best proxy for student achievement, or for overall student outcomes; I further discuss these concerns in Section 3.3.

Goe (2007) proposes a framework for conceptualizing teacher quality in which teacher qualifications, teacher characteristics, and teacher practices are inputs and teacher effectiveness is the output. Effectiveness is measured by teacher value-added, as previously described. Teacher qualifications include the teacher's educational background, certification, years of experience, and other career credentials. Teacher characteristics describe both observables such as race and gender and unobservables such as attitudes and beliefs toward teaching. Lastly, teacher practices include both in- and out-of-the-classroom activities such as lesson planning, lecturing, and student interactions. This framework can be depicted by the following model.

$$VA_{tsy} = f(q_{tsy}, c_{tsy}, p_{tsy}) \tag{2}$$

VA_{tsy} denotes the value-added score of teacher t in school s and year y, and it is a function of a vector of teacher qualifications q_{tsy}, a vector of teacher characteristics c_{tsy}, and a vector of teacher practices p_{tsy}. Various other studies offer support for this model. Egalite et al. (2015) find that teacher diversity (a component of c_{tsy}) has positive benefits on test scores of students from marginalized backgrounds, and Ayra and Kösterelioğlu (2021) find that teachers' engagement in professional development and lesson planning (a component of p_{tsy}) increases students' academic achievement. Kane et al. (2007) examine data on New York City public school teachers and find that experience (a component of q_{tsy}) matters. In comparison to students of first-year teachers, students of third-year teachers have math scores that are 6 percent of a standard deviation higher and reading scores that are 3 percent of a standard deviation higher, ceteris paribus.

Next, to extend Goe (2007)'s individual-level model of teacher quality to a school-level model of instructional quality, I average across all teachers within a school, as seen in the following equation.

$$\overline{VA}_{sy} = \overline{f(q_{sy}, c_{sy}, p_{sy})} \tag{3}$$

However, simply averaging Goe (2007)'s model of individual-level teacher attributes may be insufficient, as the literature suggests that overall instructional quality may also depend on broader institutional factors. In a study of productivity of workers at a large supermarket chain, Mas and Moretti (2009) find strong evidence of peer effects: Namely, an increase in coworker productivity of 10 percent causes individual productivity to increase 1.5 percent. It is reasonable to assume that such findings translate to the educational setting, especially as teachers who feel supported by and collaborate with their colleagues exhibit higher levels of career satisfaction, likely making them more productive (Allensworth et al., 2009). In addition, as Angrist and Lavy (1999) find, class size has a significant and negative effect on test scores, with larger classes leading to lower average scores. Moreover, assigning teachers to teach outside of their field, assigning new curriculum they are unprepared to teach, or cutting back on certain course offerings could have negative impacts on value-added that are independent of the individual teacher's attributes. Thus, I further adapt Goe (2007)'s model to incorporate these organizational effects, resulting in the model displayed below.

$$\overline{VA}_{sy} = \overline{f(q_{sy}, c_{sy}, p_{sy})} + g(O_{sy}) \tag{4}$$

In this updated model of instructional quality, average teacher value-added in a given school and year is the average of a function of individual teacher attributes plus a function of organizational effects. Under the assumption that schools seek to achieve gains in student test scores – which, given that there have historically been monetary incentives for doing so, likely holds (McArdle, 2010) – schools would then make choices to maximize this function with respect to their budget constraint, which can be represented as follows.

$$max\ \overline{VA}_{sy}\ s.t.\ C_F[r, K] + C_V[\alpha, w, n_t, \beta, n_i, \gamma, n_a] \leq B,\ C_V = (\alpha + w)n_t + \beta n_i + \gamma n_a \tag{5}$$

This constraint states that a school's total fixed costs C_F plus its total variable costs C_V must be less than or equal to its total available budget B. Fixed costs include acquisition of capital K, such as the academic building and other constant expenses, at a rental rate r. Variable costs are the sum of total teacher, student, and administrator spending. Teacher wage is denoted by w, α denotes additional per-teacher spending, and n_t denotes number of teachers. Per-student spending is denoted by β, n_i denotes number of students, γ denotes per-adminstrator spending, and n_a denotes number of administrators. C_V is an increasing function of all its parameters; that is, as the per-capita cost or total number of teachers, students, or administrators increases, so does total variable cost. Notably, while the school may have more control over some of these variables, such as hiring a certain number of teachers, other factors may be out of its control – for instance, teacher wage may be set by the district, and schools may face problems enrolling students or retaining teachers. Budget constraints may also differ between public and charter schools. Given charter schools' greater operational autonomy and lower rates of unionization, they may have more power to set wages, determine student enrollment, or otherwise alter spending decisions (Ableidinger and Hassel, 2010).

3.2 Effects of Turnover on Instructional Quality

To understand how teacher turnover might affect this model of instructional quality, I compare average value-added in year y to average value-added in year $y+1$, as shown in Equation 6, then substitute each side with the model, as shown in Equation 7. For simplicity, I assume that organizational effects remain unchanged from year y to $y+1$ and that all teachers who turn over are replaced, i.e., the total number of teachers remains constant between years. These assumptions allow the comparison to be reduced to Equation 8, and, substituting the functions for the individual teachers' averages that they represent, Equation 9. As any changes to these sums of individual teachers' attributes from year y to $y+1$ would be driven by compositional changes to the teaching staff, the key comparison is between the attributes of the departing teachers and the attributes of those who replace them, shown in Equation 10.

$$\overline{VA}_{sy} \stackrel{?}{=} \overline{VA}_{sy+1} \tag{6}$$

$$\overline{f(q_{sy}, c_{sy}, p_{sy})} + g(O_{sy}) \stackrel{?}{=} \overline{f(q_{sy+1}, c_{sy+1}, p_{sy+1})} + g(O_{sy+1}) \tag{7}$$

$$\overline{f(q_{sy}, c_{sy}, p_{sy})} \stackrel{?}{=} \overline{f(q_{sy+1}, c_{sy+1}, p_{sy+1})} \tag{8}$$

$$\sum_{t=1}^{n_y} f(q_{tsy}, c_{tsy}, p_{tsy}) \stackrel{?}{=} \sum_{t=1}^{n_{y+1}} f(q_{tsy+1}, c_{tsy+1}, p_{tsy+1}) \tag{9}$$

$$\sum_{t=1}^{n_y} \mathbb{1}(Leaver = 1) f(q_{tsy}, c_{tsy}, p_{tsy}) \stackrel{?}{=} \sum_{t=1}^{n_{y+1}} \mathbb{1}(New = 1) f(q_{tsy+1}, c_{tsy+1}, p_{tsy+1}) \tag{10}$$

The indicator function $\mathbb{1}(Leaver)$ equals 1 if the year y teacher t has left the school in year $y+1$ and 0 otherwise. The indicator function $\mathbb{1}(New)$ equals 1 if a given teacher t is new to the school in year $y+1$ and 0 otherwise. Thus, assuming organizational effects and total employment levels remain constant, the effects of turnover on instructional quality depend on whether the total value added by individual attributes of departing teachers outweighs the total value added by individual attributes of incoming teachers, in which case turnover would decrease quality.

To see this in practice, consider the simple example of total years of teaching experience, one of the determinants of q_{tsy}. I focus on this variable in particular since Ronfeldt et al. (2013), Hanushek et al. (2016), and Sorensen and Ladd (2020) all identified it as a possible mechanism for the effects of turnover. As Kane et al. (2007) found, a teacher's experience is positively correlated with their value-added. However, the relationship between teacher turnover and average total years of teaching experience – and thus between turnover and value-added – is less clear. In year y, let the average total years of teaching experience in school s be e_{sy}, computed as the sum of each teacher's total years of teaching experience, e_{ty}, divided by the total number of teachers, n_y.

$$e_{sy} = \frac{1}{n_y} \sum_{t=1}^{n_y} e_{ty} \tag{11}$$

In the subsequent year $y+1$, provided that no teachers leave and no new teachers are

hired (that is, $n_{y+1} = n_y$), the average total experience would be $e_{sy+1} = e_{sy} + 1$. However, if n teachers leave and are replaced by n new teachers, the average total experience in year $y + 1$ would then be as follows.

$$e_{sy+1} = \frac{1}{n_y}(\sum_{t=1}^{n_y} \mathbb{1}(Leaver = 0)(e_{ty} + 1) + \sum_{t=1}^{n_y} \mathbb{1}(New = 1)e_{ty+1}) \tag{12}$$

This represents average total experience as the sum of each non-leaving teacher's prior years of experience plus one, plus the sum of each new teacher's current years of experience, divided by the total number of teachers, which is assumed to be unchanged from the prior year. Importantly, this equation highlights that there is not necessarily a monotonic relationship between turnover and average years of teaching experience: If a highly experienced teacher leaves and is replaced by a teacher with no teaching experience, average experience may fall. Conversely, a less experienced teacher may be replaced by a more experienced one, albeit one who is new to the school, which could lead to an increase in average experience. These conditions can be summed up by the following inequalities.

$$\begin{aligned} &\mathbb{E}[e_{sy+1}|turnover = 0] > \mathbb{E}[e_{sy+1}|turnover > 0] \Leftrightarrow \\ &\sum_{t=1}^{n_y} \mathbb{1}(Leaver = 1)(e_{ty} + 1) > \sum_{t=1}^{n_y} \mathbb{1}(New = 1)e_{ty+1} \end{aligned} \tag{13}$$

The biconditional states that turnover will decrease average experience in the subsequent year, as opposed to no turnover, if and only if the sum of each leaving teacher's experience in year y plus one is greater than the sum of each new teacher's experience in year $y + 1$. While Sorensen and Ladd (2020) find evidence that this tends to be true, as the proportion of teachers with 3 or fewer years of experience falls as turnover rates rise, this cannot be universally claimed; the impact of turnover on teaching experience, and thereby value-added, is not definitive.

Similarly, other individual teacher attributes may also not be monotonically affected by turnover. Average teacher characteristics may change in a way that increases or decreases

teacher value-added based on the demographics of replacement teachers: If turnover causes the teaching population to become less representative of student demographics, this could harm value-added (Egalite et al., 2015). And, while certain teacher practices such as lesson planning positively affect value-added (Ayra and Köstereliogˇlu, 2021), the proportion of teachers who engage in such practices could vary based on type of turnover. If the majority of a school's turnover is among teachers with strong teaching practices who leave voluntarily, turnover may cause the quality of average teacher practices to fall, lowering average value-added as a result. However, if a school is be able to identify and selectively dismiss weaker teachers, as Barrett et al. (2020) suggest may be the case in charter schools, turnover could improve teacher quality and consequently increase average value-added.

Moreover, relaxing the assumption that organizational effects remain constant year to year, such effects may not be independent of turnover. For instance, if non-cooperative teachers are replaced by more cooperative ones, turnover may generate more positive peer effects and thus higher value-added, consistent with Barrett et al. (2020)'s idea of negative selection. Alternatively, when coworkers depart, teachers may lose people that they previously exchanged instructional tips, planned lessons, and developed friendships with; the loss of this support system may lead their productivity and overall value-added to fall. The literature provides stronger evidence in favor of this latter hypothesis, with Guin (2004) finding that schools facing higher rates of teacher turnover have more difficulty maintaining "positive working relationships" among teachers. In addition, as Ronfeldt et al. (2014) theorize, administrators may respond to significant turnover by reassigning teachers to students in a suboptimal way. Especially given nationwide teacher shortages, schools may struggle to fill vacancies left by departing teachers and compensate by assigning remaining teachers to larger class sizes or to subjects they are less qualified to teach, thus lowering value-added.

Lastly, aside from its impacts on the model of instructional quality itself, turnover may also shift a school's budget constraint. Barnes et al. (2007) find the costs of recruiting, hiring, and training new teachers to be significant: While estimates ranged from approximately $4,300 to $17,900 per teacher depending on locale, turnover imposed substantial financial

burdens on schools across the board. In Equation 5, this would be represented by an increase in α, the school's non-wage per-teacher costs. All else constant, the school would be forced to divert more resources into training new teachers and invest less in other educational areas, such as providing less training for existing teachers, employing fewer teachers and instead expanding class sizes, or cutting overhead spending. These increased costs would then reduce the maximum average value-added that a school could attain. However, as the Barnes et al. (2007) study focused solely on public schools, effects on charter schools may differ. As previously mentioned, charter schools' budget constraint may be more flexible, which could allow them to better adapt to teacher turnover and mitigate its costs.

The discussion of how turnover may impact instructional quality within a school can thus be summed up with two competing hypotheses. First, turnover may decrease average teacher value-added by diminishing the quality of average teacher attributes via loss of high-quality teachers, worsening organizational effects via institutional disruption, or raising per-teacher costs via onboarding new teachers. Alternatively, turnover may increase average teacher value-added by increasing the quality of average teacher attributes or organizational effects via the removal of lower-quality teachers. While evidence from the existing literature largely supports the former hypothesis, these conclusions primarily apply to public schools; I aim to expand upon Barrett et al. (2020) in investigating whether the same patterns hold among charters.

3.3 Modeling Student Outcomes

In examining long run outcomes, Mincer (1974) describes income as a function of years of schooling and years of work experience. This relationship is commonly depicted as in Equation 14 below, where Y represents income, S represents years of schooling, and E represents years of work experience.

$$ln(Y) = \beta_0 + \beta_1 S + \beta_2 E + \beta_3 E^2 + \varepsilon \tag{14}$$

The Mincer model typically finds a 9 to 15 percent marginal rate of return to education

(Lovenheim and Turner, 2018) and understands education as a human capital investment. However, the idea that returns to education stem from increases in human capital suggests that it is not just years of schooling but also quality of schooling that influences outcomes. Hanushek and Woessman (2010) find evidence for this: Namely, test scores are a stronger predictor of GDP per capita than years of schooling. So, I expand upon Mincer (1974)'s model by averaging the individual-level model to a school-level one and then incorporating Equation 4 to account for institutional quality, as depicted below.

$$\overline{ln(Y)}_{sy} = \beta_0 + \beta_1 \overline{S}_{sy} + \beta_2 \overline{VA}_{sy} + \beta_3 \overline{E}_{sy} + \beta_4 \overline{E^2}_{sy} + \varepsilon_{sy} \tag{15}$$

This model represents students' average long-term outcome – specifically, income – for a given school and year as an increasing function of educational quantity and quality, and a concave function with respect to years of work experience. That is, $\frac{\partial \overline{ln(Y)}}{\partial \overline{S}} > 0$, $\frac{\partial \overline{ln(Y)}}{\partial \overline{VA}} > 0$, and $\frac{\partial^2 \overline{ln(Y)}}{\partial^2 \overline{E}} < 0$.

In the short run, student outcomes would be measured from an academic standpoint, leading to an emphasis on the quantity and quality components of the updated Mincer model ($\overline{S}_{sy}$ and $\overline{VA}_{sy}$, respectively). These can be captured by a number of metrics: Attendance rates, dropout rates, and graduation rates are by definition correlated with quantity of schooling, while measures such as test scores are correlated with quality. Such short-run outcomes do tend to be good proxies of longer-run outcomes; in particular, Chetty et al. (2022) find strong support for the value-added measures described in Section 3.1. Students assigned to high value-added teachers are not only more likely to get higher test scores but are also more likely to attend college, earn more, live in neighborhoods of higher socioeconomic status, and save more for retirement. In fact, the authors estimate that a one standard deviation improvement in teacher value-added, i.e., replacing a median value-added teacher with a teacher at the 84th percentile, leads to a 1 percent increase in a student's earnings by age 28. Therefore, when modeling student outcomes, short-term outcomes can be indicative of unobserved long-term ones. And, while the model of instructional quality is not the sole determinant of student outcomes, it likely plays a significant role.

3.4 Effects of Turnover on Student Outcomes

Given this model of student outcomes described in Equation 15, I draw hypotheses about how it may be impacted by teacher turnover. I assume that, if turnover affects student outcomes, it operates solely through effects on educational quantity and quality, not work experience. As described in Section 3.2, turnover may have a positive or negative impact on educational quality, which is positively correlated with long-term student outcomes. With respect to educational quantity, which is also positively correlated with long-term student outcomes, turnover likely does not have direct effects – due to mandatory schooling laws, schools likely will not turn existing students away or cut instructional hours. However, turnover could affect quantity indirectly via quality by destabilizing the learning environment and disrupting incentives for students to attend class (affecting $g(O_{sy})$ in Equation 4). On the other hand, if quality improves via negative selection induced by turnover, quantity of education could subsequently increase, improving student outcomes.

Thus, I modify and expand upon the hypotheses from Section 3.2 as follows:

- $\mathbf{H_1}$: Turnover may harm short- and long-run student outcomes by worsening educational quality via impacts on teacher attributes or organizational effects, or by increasing per-teacher costs and thereby diverting investment from other areas.
- $\mathbf{H_2}$: Turnover may improve short- and long-run student outcomes by improving educational quality via negative selection.
- $\mathbf{H_3}$: In charter schools, negative effects of turnover on student outcomes may be mitigated due to greater operational flexibility.

Given the literature, I expect H_1 to hold over H_2. While H_3 remains largely untested, I seek to provide evidence either supporting or discrediting it.

4 Data and Methodology

4.1 Descriptive Statistics

I investigate the impact of teacher turnover on student achievement in Illinois along multiple dimensions, including ELA and math proficiency rates on standardized tests, student attendance rates, high school dropout rates, high school graduation rates, and average Scholastic Aptitude Test (SAT) math and reading scores. I do not observe long-term student outcomes, and so as described in Section 3.3 I approximate them via these short-term outcomes, all of which pertain to instructional quantity and instructional quality.

Data is drawn from the Illinois State Board of Education's Report Card Data Library, which includes annual demographic, performance, and retention data for both charter and public schools throughout the state (Illinois State Board of Education, 2024). The full dataset, which I use to evaluate effects of turnover on standardized test proficiency rates and student attendance rates, consists of 34,257 observations on the school-by-year level. Data is merged to span nine academic years, from 2014-2015 through 2022-2023, with an average of about 3806 schools in each year.

Table 1: Descriptive Statistics (Full Sample)

	Mean	SD	N
3-Year Teacher Turnover Rate	15.292	11.359	34257
Imputed Turnover	0.012	0.110	34257
Charter School	0.032	0.176	34257
% ELA Proficiency	32.829	18.568	28534
% Math Proficiency	27.157	18.580	28518
Student Attendance Rate	93.732	3.955	33508
Student Enrollment	504.640	472.593	34257
% White Students	57.511	31.886	34257
% Low-Income Students	51.704	27.828	34257

Notes: Data on all Illinois public and charter schools sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No ELA and math proficiency data from the 2019-2020 school year, as no standardized tests were administered due to the COVID-19 pandemic.

Table 1 displays descriptive statistics for this sample. The first row details the three-year

teacher turnover rate, the calculation for which is further described in Section 4.4. As 1.25 percent of observations are missing turnover data, I use simple imputation by mean and charter status, given that turnover rates – as well as missingness in turnover rates – differ significantly between charter and public schools (see Section 4.2 for further exploration of charter and public school distinctions). About 3 percent of observations are charter schools, as shown in the third row of the table. The fourth through sixth rows of the table detail the student outcome variables I use the full sample to investigate. The ELA proficiency rate measures the percent of students in a given school that are considered proficient on the ELA component of a standardized test, while the math proficiency rate measures the percent of students in a given school that are considered proficient on the math component of a standardized test. Proficiency can be achieved via a performance level of 4 or 5 on the Illinois Assessment of Readiness (IAR), a performance level of 3 or 4 on the Dynamic Learning Maps-Alternate Assessment (DLM), or a performance level of 3 or 4 on the SAT. Further details of each exam and its proficiency guidelines can be found in Table A1. Notably, as Illinois suspended standardized testing during the 2019-2020 school year due to the COVID-19 pandemic, both ELA and math proficiency rates are missing for that year. The third student outcome measure, attendance rate, describes the average percentage of days students are present out of the total number of academic days for a given school in a given year. Finally, the last three rows of the table provide an overview of key student demographics. Student enrollment measures the number of full-time students in a given school in the fall of the academic year. Percent white students measures the percentage of students who are white and non-Hispanic. Percent low-income students measures the percentage of students who live in a household with an income below the poverty line; are eligible for income-based governmental benefits; or are classified as homeless, runaway, or foster children.

Aside from this primary sample, I also examine a subsample of high schools containing 6,376 observations with an average of about 708 high schools in each year. This subsample is summarized in Table 2. As with the descriptive statistics for the full sample, the first three rows of the table display the three-year teacher turnover rate, proportion of observations for

Table 2: Descriptive Statistics (High School Sample)

	Mean	SD	N
3-Year Teacher Turnover Rate	16.099	14.920	6376
Imputed Turnover	0.024	0.153	6376
Charter School	0.087	0.282	6376
High School Dropout Rate	3.733	6.525	5772
4-Year High School Graduation Rate	85.475	13.740	6296
Average SAT Reading Score	479.559	46.568	4213
Average SAT Math Score	467.462	52.566	4213
Student Enrollment	860.876	898.197	6376
% White Students	64.253	33.131	6376
% Low-Income Students	49.651	26.599	6376

Notes: Data on all Illinois public and charter high schools sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No SAT data prior to the 2016-2017 school year, as the test did not take its current format. No SAT data in the 2019-2020 school year due to the COVID-19 pandemic.

which turnover was imputed, and proportion of charter schools. While the turnover rate is similar to that of the full sample, the fraction of imputed turnover rates and the fraction of charter schools are both higher. The following four rows display the four measures of student outcome I analyze using this subsample. The high school dropout rate of a given school and year measures the fraction of all students enrolled in the prior year who are not currently enrolled and do not meet certain exclusionary conditions (graduating, changing schools, emigrating, on a health-based leave of absence, or dying). The four-year high school graduation rate of a given school and year measures the fraction of currently graduating students out of all students who entered ninth grade for the first time four years prior, plus students who transferred into that cohort over the next three years and minus students who transferred out (changing school, emigrating, becoming imprisoned, or dying). While both of these outcomes are measured across all years of data, the remaining two – average SAT reading and math scores – are only reported beginning in 2016, as this was when the test was changed to the current 400-1600 scale. This cumulative score is the combination of scores on both the reading and math portions, each of which are measured on a 200-800 scale. SAT score data is also missing in the 2019-2020 school year due to the COVID-19 pandemic. Lastly, the bottom three rows of the table detail student demographics; while high schools

have higher student enrollment, they appear similar to the full sample in terms of student race and socioeconomic status.

4.2 Descriptive Statistics by Charter Status

As a key portion of my analysis involves examining heterogeneity between public and charter schools, I also present descriptive statistics for the full sample split by charter status in Table 3. Following the literature, the charter schools in my sample exhibit significantly higher three-year turnover rates. The charter schools also display significantly poorer student outcomes, as measured by both standardized test proficiency rates and attendance rates. This may be in part due to differences in student body composition between public schools and charter schools in Illinois. While approximately 58.4 percent of public school students in the data are white and approximately 50.6 percent are low-income, these numbers are 30.5 and 85.1 in charter schools. Thus, charter schools' lower average outcomes may be explained by their higher percentage of students from disadvantaged backgrounds. I present similar statistics for the sample of high schools in Table 4 and find similar results: significantly higher turnover among charter schools, as well as significantly poorer student outcomes (lower graduation rates, higher dropout rates, and lower average SAT scores) and differences in demographics.

Table 3: Descriptive Statistics by Charter Status (Full Sample)

	Public			**Charter**			**T-Tests**	
	Mean	SD	N	Mean	SD	N	Difference	p-Value
3-Year Teacher Turnover Rate	14.263	8.733	33162	46.461	26.908	1095	-32.199***	0.000
Imputed Turnover	0.005	0.071	33162	0.232	0.422	1095	-0.227	0.000
% ELA Proficiency	33.387	18.466	27617	16.023	12.916	917	17.364***	0.000
% Math Proficiency	27.671	18.548	27602	11.692	11.587	916	15.979***	0.000
Student Attendance Rate	93.928	3.271	32432	87.820	11.343	1076	6.109***	0.000
Student Enrollment	504.215	462.363	33162	517.528	716.445	1095	-13.313	0.359
% White Students	58.404	31.637	33162	30.458	27.213	1095	27.946***	0.000
% Low-Income Students	50.601	27.424	33162	85.126	17.248	1095	-34.526***	0.000

Notes: Data on all Illinois public and charter schools sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No ELA and math proficiency data from the 2019-2020 school year, as no standardized tests were administered due to the COVID-19 pandemic.

Table 4: Descriptive Statistics by Charter Status (High School Sample)

	Public			Charter			T-Tests	
	Mean	SD	N	Mean	SD	N	Difference	p-Value
3-Year Teacher Turnover Rate	13.410	9.783	5822	44.358	26.127	554	-30.948***	0.000
Imputed Turnover	0.005	0.068	5822	0.227	0.420	554	-0.223***	0.000
4-Year High School Graduation Rate	86.667	11.189	5759	72.686	26.332	537	13.982***	0.000
High School Dropout Rate	2.947	3.561	5288	12.319	17.009	484	-9.372***	0.000
Average SAT Reading Score	484.854	44.082	3812	429.218	38.902	401	55.636***	0.000
Average SAT Math Score	472.889	50.203	3812	415.872	46.211	401	57.017***	0.000
Student Enrollment	887.268	885.344	5822	583.529	982.408	554	303.738***	0.000
% White Students	67.514	31.765	5822	29.980	27.273	554	37.534***	0.000
% Low-Income Students	46.039	24.617	5822	87.604	14.033	554	-41.565***	0.000

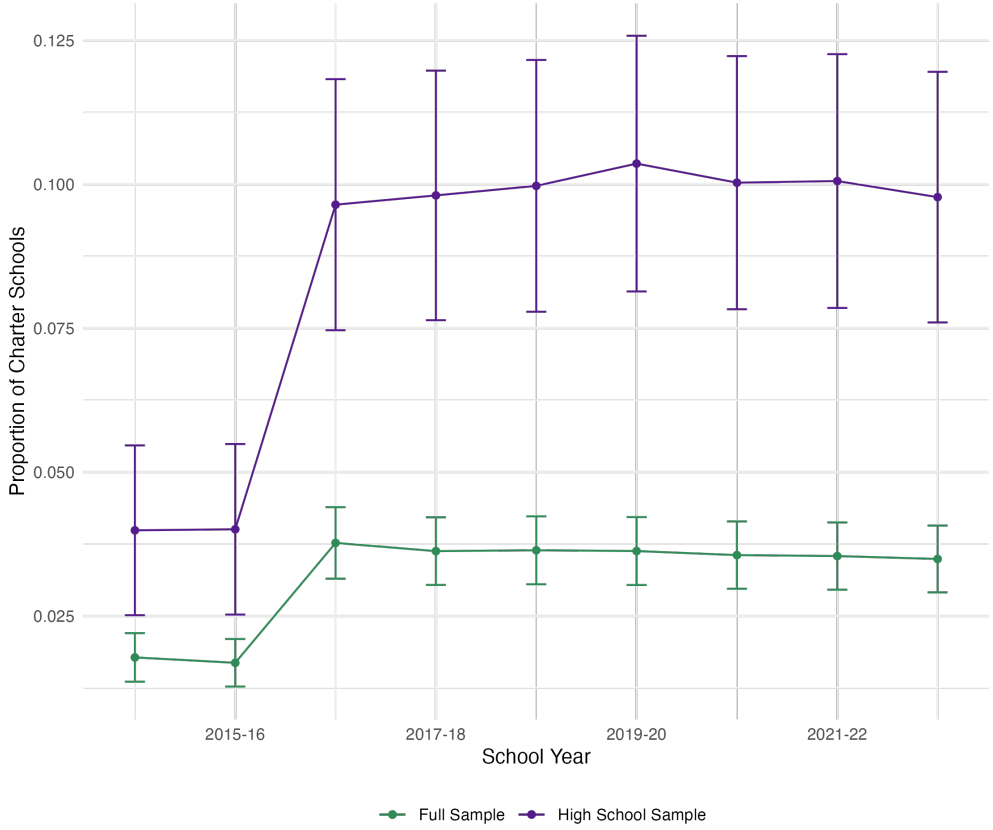
Notes: Data on all Illinois public and charter high schools sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No SAT data prior to the 2016-2017 school year, as the test did not take its current format. No SAT data in the 2019-2020 school year due to the COVID-19 pandemic.

4.3 Time Trends

In keeping with the literature, turnover rates in the dataset remain relatively stable over time and are significantly higher among charter schools. This holds both for the full sample and for the high school sample, as shown in Figure A1. Key outcome variables for the full sample – percent ELA proficiency, percent math proficiency, and student attendance rate – also generally do not show much variation over time, though they do exhibit a significant dip following the 2019-2020 school year, as seen in Figure A2. Such a dip occurs among both public and charter schools and is consistent with the numerous studies suggesting the COVID-19 pandemic and resulting school shutdowns, transition to remote learning, and other academic disruptions had a negative and lasting impact on student outcomes (Kuhfeld et al., 2022). Outcomes measured within the high school sample – four-year graduation rate, dropout rate, and average SAT reading and math scores – are less observably affected by the pandemic. As depicted in Figure A3, four-year graduation rates remain fairly consistent over time, while average SAT reading and math scores trend slightly downward. Dropout rates among public high schools trend slightly upward, as do dropout rates among charter high schools. However, dropout rates in charters spike in the 2021-2022 school year; given that this outcome was measured following the end of the first full year of primarily remote learning in Illinois, which disproportionately affected disadvantaged students (Swartz, 2021), the spike

could be attributed to effects of the pandemic. More generally, as previously observed, charter schools tend to have lower student outcomes likely due to their differing student population.

Figure 4: Proportion of charter schools in Illinois over time



Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years.

Another crucial time trend to note is the sharp increase in the proportion of charter schools between the 2015-2016 school year and the 2016-2017 school year, depicted in Figure 4. This jump was driven by an apparent increase in the number of charter schools from 63 to 136 in the full sample and from 27 to 68 in the high school subsample; the number of public schools remained relatively steady. However, this is not indicative of new charter schools opening but rather a reclassification in the data reporting of the Illinois State Board of Education. Starting in the 2016-2017 school year, in an attempt to increase transparency

surrounding charter schools, the Board began reporting data on the individual campus level; previously, charter schools operating under the same charter contract but on multiple campuses had been reported as one entity (Smith, 2018). This explanation is supported by an examination of student enrollment, which, as seen in Figure A4, remains relatively constant across time for both public and charter schools in the full sample. Still, enrollment does dip in charter schools in the high school subsample following the 2015-2016 school year, likely in part due to the Board's closure of eight charter schools (Smith, 2018).

4.4 Empirical Approach

Naslund and Ponomariov (2018) apply an ordinary least squares (OLS) regression with interaction terms to measure the differential effects of teacher turnover in charter as compared to public schools. Yet, as their analysis is limited to cross-sectional district-level data, such an approach may not be exogenous: They control for a variety of district-level demographics such as enrollment, average teacher salary, and percent of white students, but it is difficult to claim that there are no omitted school-level variables that may bias the results. Thus, while I retain the interaction terms from their model, I build upon it by using panel data on the school and year level. Following Ronfeldt et al. (2013), Hanushek et al. (2016), and Sorensen and Ladd (2020), I use a fixed effects model that controls for unobservables that vary at the school level and at the year level. The critical assumption to this approach is that year fixed effects are constant across schools, and similarly, school fixed effects are constant across years.

The initial fixed effects regression model is as follows in Equation 16. The dependent variable $outcome_{it}$ denotes the average student outcome of school i in year t, with $outcome \in$ {ELA proficiency rate, math proficiency rate, attendance rate} for the full sample and $outcome \in$ {graduation rate, dropout rate, average SAT reading score, average SAT math score} for the high school sample. The covariate $turnover_{it}$ denotes the three-year teacher turnover rate of school i in year t; in keeping with Sorensen and Ladd (2020), I use a longer-term measure of instability as opposed to single-year turnover in hopes of better capturing its full effects. Further details of this calculation are given by Equation 17; *returning* is

defined by the number of teachers in a given school and year who taught at the school the prior year, and *total* is the total number of teachers employed at a given school in a given year. Returning to the discussion of Equation 16, the interaction term $charter_i * turnover_{it}$ is the interaction between school i's charter status and its three-year teacher turnover rate. School fixed effects are denoted by δ_i, and year fixed effects are denoted by λ_t.

The coefficients of interest are those of the turnover coefficient and the interaction term, β_1 and β_2. While β_1 can be expected to be negative if turnover generally harms student outcomes, the interaction term coefficient would capture the differential effects of turnover in charter as compared to public schools. For instance, a positive value of β_2, combined with a negative β_1, would indicate that the detrimental effects of turnover are lower in charter schools than in public schools. To determine whether this sum is significant, I compute the standard error of the sum, displayed in Equation 18, and use it to construct a 95 percent confidence interval around $\beta_1 + \beta_2$.

$$outcome_{it} = \alpha + \beta_1 turnover_{it} + \beta_2 charter_i * turnover_{it} + \delta_i + \lambda_t + \varepsilon_{it} \tag{16}$$

$$turnover_{it} = 1 - \frac{returning_{it} + returning_{it-1} + returning_{it-2}}{total_{it} + total_{it-1} + total_{it-2}} \tag{17}$$

$$SE(\beta_1 + \beta_2) = \sqrt{SE(\beta_1)^2 + SE(\beta_2)^2 + 2 * Cov(\beta_1, \beta_2)} \tag{18}$$

Given that there are upwards of 4,000 schools in the full sample and upwards of 760 schools in the high school sample, it may be computationally inefficient to create dummy variables for each school. Thus, to improve upon the model in Equation 16, I demean the fixed effects, beginning by averaging over t for each i as shown in Equation 19.

$$\overline{outcome_i} = \alpha + \beta_1 \overline{turnover_i} + \beta_2 charter_i * \overline{turnover_i} + \delta_i + \overline{\lambda} + \overline{\varepsilon_i} \tag{19}$$

To obtain the final form of the fixed effects model, I subtract Equation 19 from Equation 16, resulting in Equation 20 below; note that the individual fixed effects have been differenced

away, allowing for a simple OLS model regressing the deviation of student outcome in a given year from the individual mean on the deviation of each of the time-based covariates from their individual means. Essentially, this process removes averages at the school level over time, and β_1 and β_2 remain unchanged from the initial dummy variable fixed effects model in Equation 16. I also cluster standard errors on the school level to account for any nonzero covariances between schools. While the fixed effects address differences in means, clustering addresses the possibility of moving averages.

$$\begin{aligned}(outcome_{it} - \overline{outcome_i}) &= \beta_1(turnover_{it} - \overline{turnover_i}) \\ &+ \beta_2 charter_i * (turnover_{it} - \overline{turnover_i}) + (\lambda_t - \overline{\lambda}) + (\varepsilon_{it} - \overline{\varepsilon_i})\end{aligned} \tag{20}$$

After running this fixed effects regression for the seven student outcome measures of interest and identifying the marginal effects of turnover given by β_1 and β_2, I compute overall turnover effects by multiplying these marginal effects by the expected value of turnover, as shown in the basic calculations given by Expressions 21 and 22.

$$(\beta_1 + \beta_2) * E[turnover|charter = 1] \tag{21}$$

$$\beta_1 * E[turnover|charter = 0] \tag{22}$$

While Expression 21 represents the expected total impact of teacher turnover in charter schools, Expression 22 represents the expected total impact of teacher turnover in public schools. These overall calculations are crucial because, while the marginal impact of turnover in charter schools may be different from that of public schools (i.e., β_2 is nonzero), the higher rates of turnover in charter schools – reflected in both the literature and in the descriptive statistics – may offset this. A statistically significant difference between Expressions 21 and 22 would indicate that, on the whole, charter school students and public school students are affected differently by the cumulative effects of teacher turnover.

5 Results

Below, I present my empirical findings. In Sections 5.1 and 5.2, I discuss the results – displayed in Table 5 – of the specifications outlined in Section 4.4 for the full sample and high school sample outcomes, respectively. Section 5.3 discusses additional specifications. Section 5.4 explores heterogeneity in the impact of turnover on ELA and math proficiency rates by various student demographics, including gender, race, and socioeconomic status. Section 5.5 further interprets the results and their limitations.

Table 5: Effects of Turnover on Student Outcomes

	(1) ELA Proficiency	(2) Math Proficiency	(3) Attendance Rate	(4) Dropout Rate	(5) 4-Year Grad. Rate	(6) Avg. SAT Reading	(7) Avg. SAT Math
3-Year Turnover Rate	-0.046***	-0.034***	0.005	-0.012	-0.010	-0.106*	-0.069
	(0.013)	(0.011)	(0.003)	(0.017)	(0.017)	(0.058)	(0.065)
Charter x Turnover	0.063***	0.069***	0.004	0.084***	-0.035	0.158**	0.221**
	(0.019)	(0.020)	(0.008)	(0.027)	(0.041)	(0.079)	(0.100)
Constant	36.383***	26.561***	94.297***	2.849***	84.543***	495.074***	484.714***
	(0.251)	(0.222)	(0.057)	(0.301)	(0.509)	(0.958)	(1.116)
School Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	28534	28518	33508	5772	6296	4213	4213

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No SAT data prior to the 2016-2017 school year, as the test did not take its current format. No ELA proficiency, math proficiency, or SAT data from the 2019-2020 school year, as no standardized tests were administered. Standard errors are clustered by school.

5.1 Full Sample Outcomes

5.1 ELA Proficiency

As seen in column (1) of Table 5, teacher turnover has a significant and negative effect on the percentage of ELA-proficient students in a given Illinois public school. Specifically, an increase in the three-year teacher turnover rate from 0 percent to 100 percent decreases the ELA proficiency rate in a public school by 4.60 percentage points, or 0.249 of a standard deviation. This effect is offset in charter schools, as the coefficient on the interaction between charter and turnover is positive and significant. Applying Equation 18 to compute the standard error of the sum of coefficients, the effect of turnover on ELA proficiency rates in charter schools is insignificant (see Table A2 for the exact estimates). On the whole,

applying Expressions 21 and 22 from Section 4.4, the cumulative effect of average turnover is a decrease in ELA proficiency of about 0.656 percentage points in public schools (0.036 SD), while I cannot conclude a statistically significant effect in charter schools.

5.2 Math Proficiency

Similarly, as seen in column (2) of Table 5, teacher turnover has a significant and negative effect on the percentage of math-proficient students in a given Illinois public school. An increase in the three-year teacher turnover rate from 0 percent to 100 percent decreases the math proficiency rate in a public school by 3.40 percentage points, or 0.183 of a standard deviation. As with ELA proficiency rates, this effect is negated in charter schools, as the coefficient on the interaction term is positive and significant, rendering the sum of the turnover and interaction term coefficients insignificant (see Table A2). Notably, turnover appears to have a smaller effect on math proficiency than ELA proficiency, contrasting the literature suggesting that math test scores tend to be more responsive to changes in the classroom environment (Anderson, 2022); however, this difference is not statistically significant. The cumulative effect of average turnover is a decrease in math proficiency rates by about 0.485 percentage points in public schools (0.026 SD). This differs from the impact of total turnover in charter schools, where I cannot conclude any significant effects.

5.3 Student Attendance Rate

I do not find any significant effects of turnover on student attendance rates, either in public or in charter schools. This may suggest that day-to-day quantity of schooling is independent of turnover, or that there are multiple effects of turnover on daily attendance that move in opposite directions so as to cancel one another out.

5.2 High School Sample Outcomes

5.1 High School Dropout Rate

While teacher turnover has no effect on dropout rates in public high schools, it has a significant and positive effect in charter high schools, as displayed in column (4) of Table 5.

That is, when the three-year teacher turnover rate in a charter high school increases from 0 percent to 100 percent, the student dropout rate increases by 8.436 percentage points, or 0.496 of a standard deviation. The cumulative effect of average turnover on charter high school dropout rates is an increase of about 3.742 percentage points (0.220 SD).

5.2 4-Year High School Graduation Rate

I do not find any significant effects of turnover on 4-year high school graduation rates, either in public or in charter schools. This is in line with Naslund and Ponomariov (2018), who also found a null effect.

5.3 Average SAT Reading Score

As shown in column (6) of Table 5, I find that teacher turnover significantly decreases the average SAT reading score in public high schools. When the three-year teacher turnover rate changes from 0 to 100 percent, average score falls by 10.638 points, or 0.241 of a standard deviation. Again, this effect is absent in charter high schools, where the coefficient on the interaction term is significant and positive, canceling out the initial turnover coefficient and resulting in a null effect. The cumulative effect of average turnover on average SAT reading scores in public high schools is a decrease of about 1.426 points (0.032 SD).

5.4 Average SAT Math Score

Again in contrast to the literature, the average SAT math score is less responsive to turnover than the average SAT reading score. As shown in the final column of Table 5, turnover has no impact on the average SAT math score of public high schools. Meanwhile, in charter high schools, turnover has a significant and positive effect: When the three-year teacher turnover rate changes from 0 to 100 percent, average score rises by 22.143 points, or 0.479 of a standard deviation. The cumulative effect of average turnover is an increase in average SAT math score of about 9.822 points (0.213 SD).

5.3 Additional Specifications

As a robustness check, I also present estimations for each outcome using only non-imputed data in Table A10; coefficients generally have similar magnitude and direction. And, aside from solely estimating the fixed effects regression for each outcome, I run six simpler specifications building up to the final model. Results of these specifications for each outcome can be found in Tables A3 through A9. Three of the additional specifications are simple OLS regressions: a regression of the relevant outcome on three-year teacher turnover; a regression of the relevant outcome on turnover and a charter school dummy; and a regression of the relevant outcome on turnover, a charter school dummy, and their interaction. Two specifications use only school fixed effects: a regression of the relevant outcome on turnover with school fixed effects and a regression of the relevant outcome on turnover, a charter and turnover interaction term, and school fixed effects. The sixth specification regresses the relevant outcome on turnover, school fixed effects, and year fixed effects. The seventh and final column of each extended results table displays the estimates for the final fixed effects model shown in Table 5. Notably, estimates vary significantly based on inclusion of fixed effects, suggesting that a large part of any initial observed correlation between turnover and student outcomes can be explained by school- and year-level factors.

5.4 Heterogeneity by Demographics

Given that the literature suggests turnover differentially affects students by demographic group, with the most disadvantaged students being the most negatively impacted (Ronfeldt et al., 2013), I conduct heterogeneity analyses. In particular, I examine the effects of turnover on ELA and math proficiency rates for male, female, white, Black, Latino, Asian, and low-income students. I do not examine heterogeneous treatment effects with regards to the remaining outcomes due to lack of by-demographic outcome data.

5.1 ELA Proficiency

Table 6 details heterogeneity in the effect of turnover on ELA proficiency rates. Comparing columns (2) and (3), there appear to be no significant differences in the effect by gender. Columns (4) through (7) display estimates broken down by racial group; notably, while ELA proficiency rates for white and Black public school students are both negatively impacted by turnover to approximately the same degree as the overall sample, there is a null effect among Latino and Asian public school students. Moreover, there is a positive impact of turnover on ELA proficiency rates among Latino and Asian charter school students, as opposed to a negative impact on Black students and a null effect on white students. Finally, the results for low-income students, depicted in column (8), do not reveal any heterogeneity by socioeconomic status.

Table 6: Heterogeneous Effects of Turnover on ELA Proficiency

	(1) *Overall*	(2) *Male*	(3) *Female*	(4) *White*	(5) *Black*	(6) *Latino*	(7) *Asian*	(8) *Low-Income*
3-Year Turnover Rate	-0.046***	-0.050***	-0.047***	-0.053***	-0.034**	-0.013	-0.033	-0.055***
	(0.013)	(0.013)	(0.014)	(0.017)	(0.017)	(0.022)	(0.049)	(0.012)
Charter x Turnover	0.063***	0.069***	0.071***	0.113**	0.039	0.048*	0.233***	0.076***
	(0.019)	(0.019)	(0.022)	(0.045)	(0.024)	(0.029)	(0.077)	(0.018)
Constant	36.383***	30.758***	42.472***	44.439***	21.745***	30.431***	65.213***	26.720***
	(0.251)	(0.259)	(0.283)	(0.316)	(0.350)	(0.380)	(0.756)	(0.255)
School Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	28534	28315	28265	23578	14388	17872	7372	27364

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No data from the 2019-2020 school year, as no standardized tests were administered. Analysis uses the full sample. Standard errors are clustered by school.

5.2 Math Proficiency

Table 7 details heterogeneity in the effect of turnover on math proficiency rates. There is no evidence of heterogeneity by gender, or by socioeconomic status. However, as with ELA proficiency rates, results vary by racial group. In public schools, while turnover harms math proficiency rates of white students, it has no impact on proficiency rates for other racial

groups. In charter schools, turnover has a positive effect on the math proficiency rates of Latino and Asian students but no effect on Black and white students.

Table 7: Heterogeneous Effects of Turnover on Math Proficiency

	(1) *Overall*	(2) *Male*	(3) *Female*	(4) *White*	(5) *Black*	(6) *Latino*	(7) *Asian*	(8) *Low-Income*
3-Year Turnover Rate	-0.034***	-0.039***	-0.031**	-0.050***	-0.012	-0.009	-0.022	-0.043***
	(0.011)	(0.013)	(0.012)	(0.015)	(0.014)	(0.020)	(0.041)	(0.011)
Charter x Turnover	0.069***	0.086***	0.064***	0.115*	0.030	0.083**	0.298***	0.080***
	(0.020)	(0.021)	(0.021)	(0.059)	(0.020)	(0.033)	(0.067)	(0.020)
Constant	26.561***	26.424***	27.046***	33.675***	12.379***	21.249***	59.631***	18.094***
	(0.222)	(0.243)	(0.239)	(0.278)	(0.280)	(0.345)	(0.703)	(0.218)
School Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	28518	28268	28240	23557	14353	17848	7375	27281

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No data from the 2019-2020 school year, as no standardized tests were administered. Analysis uses the full sample. Standard errors are clustered by school.

5.5 Interpretations and Limitations

Overall, I find significant negative impacts of turnover on standardized test proficiency rates in public schools, driven primarily by effects on Black and white students. For the most part, these results are consistent with the findings of Ronfeldt et al. (2013), Hanushek et al. (2016), and Sorensen and Ladd (2020), though a key difference is that I find no heterogeneity by socioeconomic status; the prior literature suggests that more economically disadvantaged students may be more affected. Another discrepancy with the literature is that, in terms of standard deviations, my results are about twice the magnitude of those previously found – a change from 0 percent teacher turnover to 100 percent teacher turnover would reduce standardized test proficiency rates by 0.249 of a standard deviation in ELA and by 0.183 of a standard deviation in math. However, these estimates may not be perfectly comparable with preexisting ones on two grounds: First, while past literature measures average changes among individual students' scores, I measure changes in school averages. Additionally, I examine proficiency rates, which measure the proportion of students in a school above and

below a certain benchmark, while past literature examines continuous test scores. In charter schools, I find no detrimental effects of turnover on proficiency rates, and Latino and Asian students even appear to benefit. This is a more positive result than that of Naslund and Ponomariov (2018), who identify negative impacts of turnover on standardized test pass rates in charter school organizations, albeit to a lesser extent than in public school districts.

With regards to other measures of academic quality, I find evidence that turnover decreases average SAT reading scores in public high schools while increasing average SAT math scores in charter high schools. In comparison to the past literature, the effect in public schools – a decrease in average reading scores by 0.241 of a standard deviation when turnover increases from 0 to 100 percent – is much larger in magnitude, though it still may not be perfectly comparable based on the different averaging method. The effect in charter schools – an increase in average math scores by 0.479 of a standard deviation when turnover increases from 0 to 100 percent – is also an economically meaningful effect, though it lacks a direct reference point in the literature. Both of these findings are directionally in line with the findings for proficiency rates, though it is noteworthy that the significance of the effects varies between the two subjects. Yet, when conducting the analysis without the imputed data, as displayed in Table A10, turnover significantly decreases average SAT scores for both subjects in public schools while having no significant effect on average scores for either subject in charter schools, suggesting the discrepancies may be a figment of the data.

Effects of turnover on student outcomes related to quantity of education are less consistent. Reinforcing the findings of Naslund and Ponomariov (2018), I identify no significant impacts of turnover on 4-year high school graduation rates in both public and charter schools. I also find a null effect on student attendance rates. However, turnover does significantly increase the high school dropout rate in charter schools, suggesting that turnover may instill some form of longer-term instability in students' quantity of education despite the lack of changes to immediate measures such as attendance rates. This is in contrast to the theory, which predicts the negative impacts of turnover to be higher among public schools than among charters. Furthermore, the effect of turnover on dropout rates is by magnitude

the largest effect I identify: When turnover rates increase from 0 to 100 percent, charter high school dropout rates jump by nearly half a standard deviation, suggesting significant real-world impacts.

Still, while the results are promising, there are also a number of limitations to their interpretation. Firstly, the regressions do not test the mechanisms behind any estimated causal effects. The findings identify the overall impact of turnover on student outcomes and provide potential support for all three hypotheses outlined in Section 3.4: Turnover may harm student outcomes by worsening educational quality or increasing a school's per-teacher costs, some of these negative impacts may be mitigated in charter schools, and turnover may in fact improve student outcomes – at least in charter schools – via some form of negative selection. But, the findings cannot be directly attributed to specific staffing or organizational changes. Given that the student and teacher populations of charter schools tend to differ from those of public schools, selection into certain school types also poses interpretive limits. Consequently, while these results measure the effects of turnover on charter and public school populations, these populations are assumed to be a given. That is, one could not conclude that a public school population randomly selected to be moved into a charter school environment would be impacted as described in the charter school results, and vice versa. Lastly, a concern with the regression strategy itself is that it may suffer from reverse causality if outcomes in a given year influence teacher turnover in a subsequent year. With the given data, it is difficult to conclusively determine whether or not this significantly biases regression results; I discuss potential improvements in Section 6 below.

6 Conclusion

In light of U.S. teachers' declining rates of job satisfaction and other alarming occupational trends nationwide, it is crucial to understand the impacts that teacher turnover – a perpetual consequence of these issues – has on student outcomes, both in traditional public schools and in the increasingly popular charter schools. I present evidence that supports and expands upon the existing literature, identifying the effects of turnover as largely detri-

mental to multiple dimensions of student outcomes. While some effects of turnover may be mitigated and even at times positive in charter schools, others, such as turnover-driven increases in high school dropout rates, appear unique to charter schools. Thus, on the whole, my results provide support for policies that aim to reduce teacher turnover across the board, whether by offering more competitive wages and benefits or by fostering a more supportive working environment and boosting individual morale (Allensworth et al., 2009). Moreover, they provide tentative evidence to both favor expansions of charter school operations and to be cautious of such expansions: The effects of turnover do differ between public and charter schools, but not in purely one-sided ways.

Given these findings, as well as the limitations of my research discussed in Section 5.5, there are a number of avenues left to explore. Replicating the analysis using more granular, individual-level student outcomes data, as opposed to school-level averages, would make a stronger case for the exogeneity of the regression and allow for additional heterogeneity analyses. Data on long-term student outcomes, such as income and savings, would remove the need to extrapolate from short-term measures. And, a more thorough investigation of the mechanisms behind turnover would help shape a more comprehensive understanding of it. Such an investigation could include examinations of how various teacher attributes, such as the fraction of novice teachers or average years of teacher experience, change in response to turnover in both public and charter schools. Another component could focus on differential effects of voluntary as compared to involuntary turnover on student outcomes as well as on a school's average instructional quality. Further research along these lines would build upon my conclusions by not just advocating for policies that reduce turnover but also promoting policies that can target specific types of turnover, more effectively address the mechanisms through which it operates, and better support teachers nationwide.

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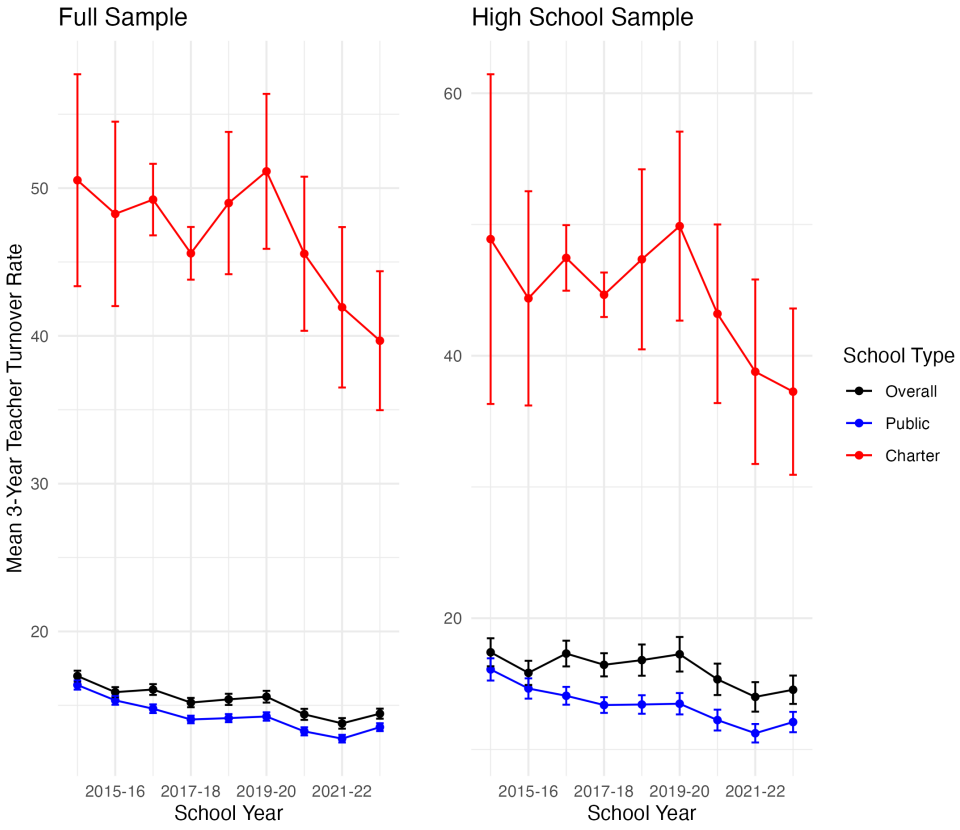
8 Appendices

Table A1: Standardized tests and proficiency benchmarks

Exam	Description	Level	Proficient?
DLM	The Dynamic Learning Maps-Alternate Assessment is the required state assessment for students in grades 3 to 11 with significant cognitive disabilities.	4: Advanced 3: At target 2: Approaching target 1: Emerging	Yes Yes No No
IAR	The Illinois Assessment of Readiness is the state assessment and accountability measure for all Illinois students enrolled in a public school district.	5: Exceeded expectations 4: Met expectations 3: Approached expectations 2: Partially met expectations 1: Did not meet expectations	Yes Yes No No
SAT	The SAT is a standardized test generally taken by students in grade 11 and widely used for college admissions in the United States.	4: Exceeds standards 3: Meets standards 2: Approaching standards 1: Partially meets standards	Yes Yes No No

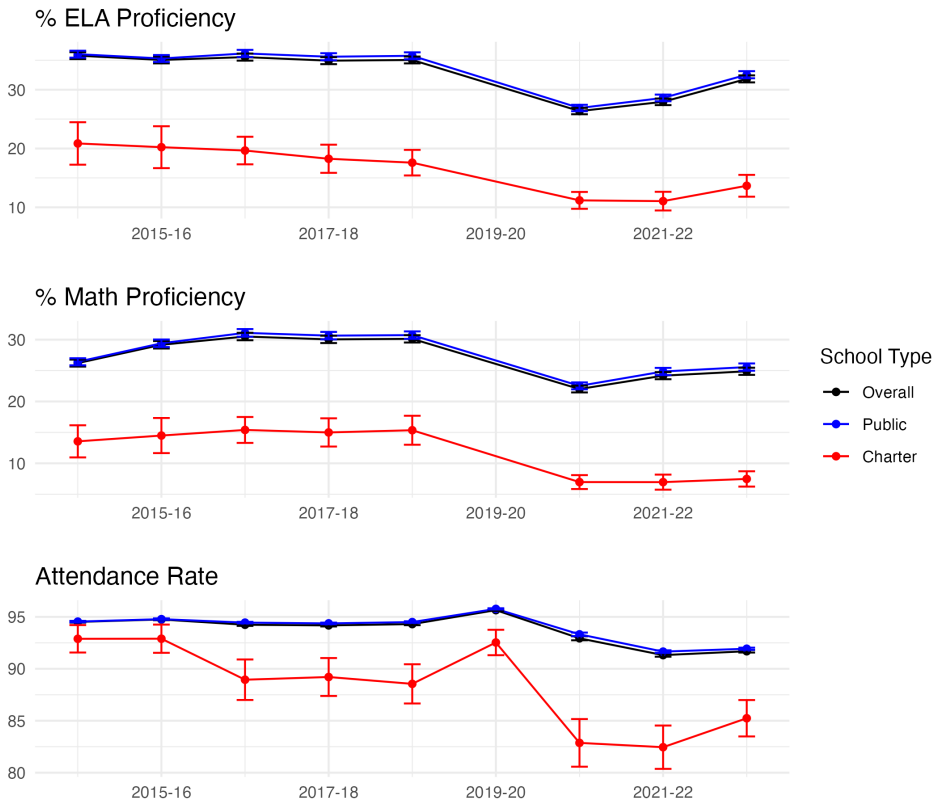
Source: Illinois State Board of Education (2024)

Figure A1: Turnover rates among Illinois schools over time



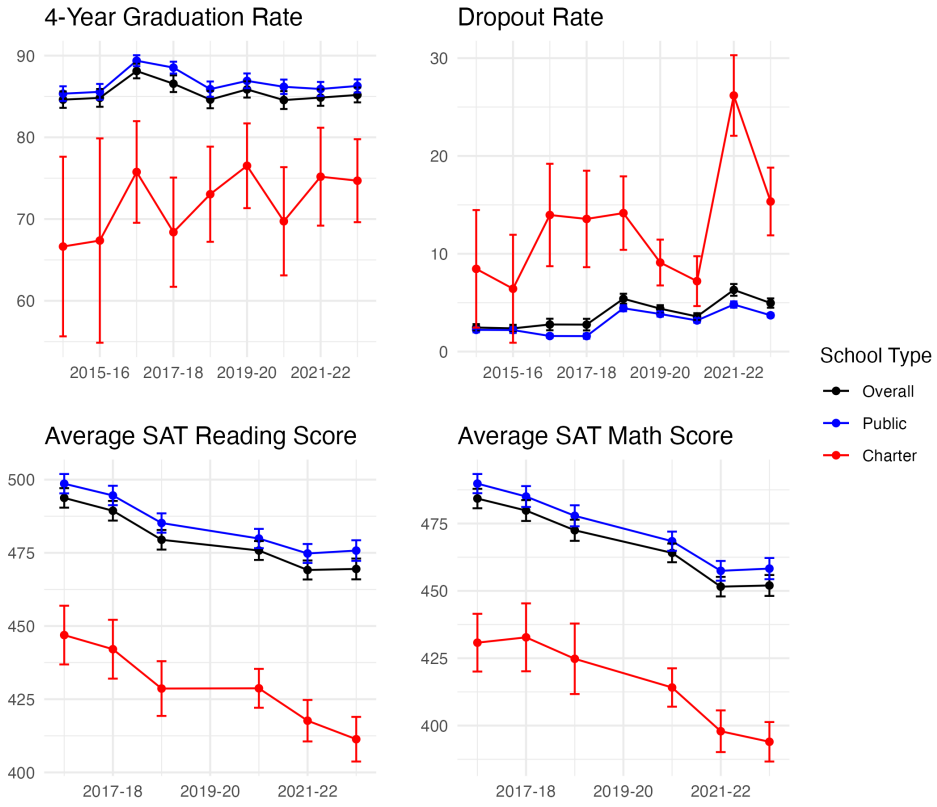
Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years.

Figure A2: Student outcomes over time (full sample)



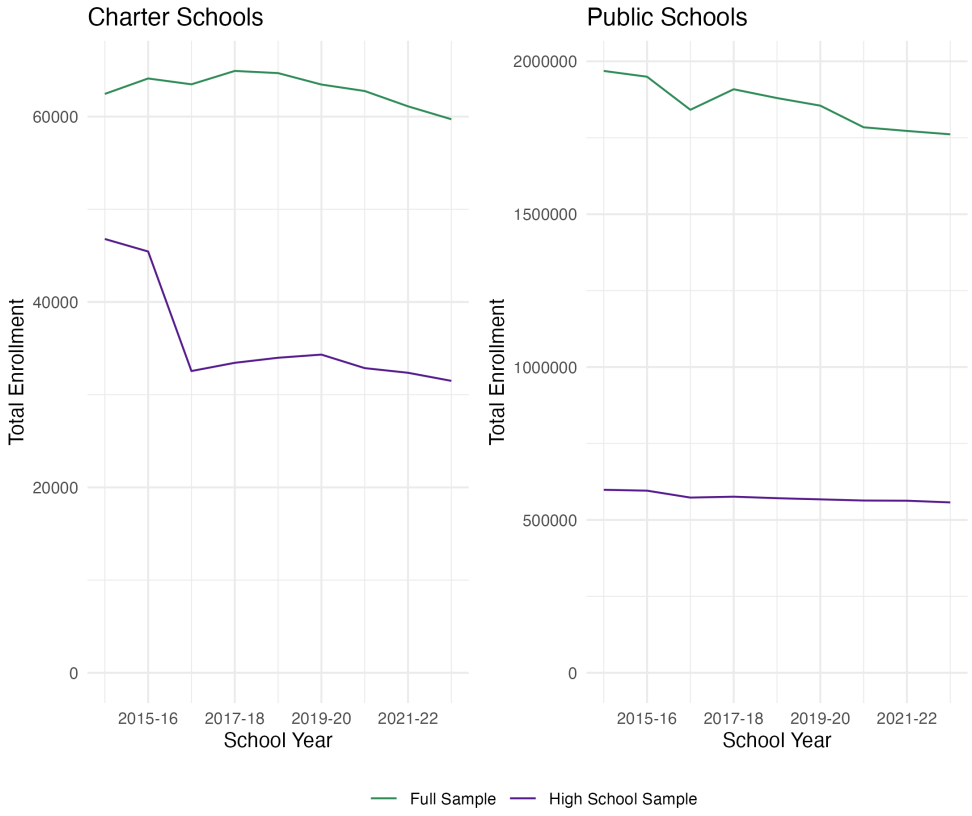
Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No ELA or math proficiency data from the 2019-2020 school year, as no standardized tests were administered.

Figure A3: Student outcomes over time (high school sample)



Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No SAT data prior to the 2016-2017 school year, as the test did not take its current format. No SAT data from the 2019-2020 school year, as no standardized tests were administered.

Figure A4: Student enrollment over time



Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years.

Table A2: Coefficient Significance in Charter Schools

Variable	$\beta_1 + \beta_2$	$SE(\beta_1 + \beta_2)$	95% CI	Significant?
% ELA Proficiency	0.017	0.076	(-0.132, 0.166)	No
% Math Proficiency	0.035	0.075	(-0.113, 0.183)	No
Average SAT Reading Score	0.051	0.172	(-0.286, 0.389)	No

Notes: To determine whether turnover has a statistically significant impact on student outcomes in charter schools, this table presents the sum $\beta_1 + \beta_2$, the standard error of this sum, and the 95% confidence interval around it for all outcome variables in Table 5 that had statistically significant values for both β_1 and β_2.

Table A3: Effects of Turnover on % ELA Proficiency

	No FEs			**School FEs**		**School and Year FEs**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
3-Year Turnover Rate	-0.451***	-0.422***	-0.541***	0.047***	0.050***	-0.029***	-0.046***
	(0.012)	(0.014)	(0.017)	(0.011)	(0.014)	(0.010)	(0.013)
Charter School		-4.115***	-23.149***				
		(0.744)	(0.974)				
Charter x Turnover			0.501***		-0.012		0.063***
			(0.023)		(0.022)		(0.019)
Constant	39.684***	39.360***	41.039***	32.058***	32.027***	36.181***	36.383***
	(0.204)	(0.219)	(0.266)	(0.167)	(0.188)	(0.230)	(0.251)
Observations	28534	28534	28534	28534	28534	28534	28534

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No data from the 2019-2020 school year, as no standardized tests were administered. Analysis uses the full sample. Columns (1) to (3) use robust standard errors while standard errors for columns (4) to (7) are clustered by school.

Table A4: Effects of Turnover on % Math Proficiency

	No FEs			**School FEs**		**School and Year FEs**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
3-Year Turnover Rate	-0.436***	-0.416***	-0.535***	0.029***	0.018	-0.015	-0.034***
	(0.011)	(0.013)	(0.017)	(0.011)	(0.013)	(0.009)	(0.011)
Charter School		-2.977***	-22.002***				
		(0.698)	(0.856)				
Charter x Turnover			0.501***		0.040*		0.069***
			(0.022)		(0.024)		(0.020)
Constant	33.784***	33.551***	35.230***	26.667***	26.771***	26.339***	26.561***
	(0.203)	(0.218)	(0.264)	(0.164)	(0.174)	(0.211)	(0.222)
Observations	28518	28518	28518	28518	28518	28518	28518

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No data from the 2019-2020 school year, as no standardized tests were administered. Analysis uses the full sample. Columns (1) to (3) use robust standard errors while standard errors for columns (4) to (7) are clustered by school.

Table A5: Effects of Turnover on Student Attendance

	No FEs			**School FEs**		**School and Year FEs**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
3-Year Turnover Rate	-0.062***	-0.024***	-0.049***	0.031***	0.034***	0.006*	0.005
	(0.003)	(0.004)	(0.003)	(0.004)	(0.004)	(0.003)	(0.003)
Charter School		-5.386***	-9.182***				
		(0.409)	(0.710)				
Charter x Turnover			0.099***		-0.010		0.004
			(0.011)		(0.010)		(0.008)
Constant	94.704***	94.268***	94.618***	93.231***	93.208***	94.284***	94.297***
	(0.049)	(0.055)	(0.048)	(0.058)	(0.057)	(0.058)	(0.057)
Observations	33508	33508	33508	33508	33508	33508	33508

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. Analysis uses the full sample. Columns (1) to (3) use robust standard errors while standard errors for columns (4) to (7) are clustered by school.

Table A6: Effects of Turnover on High School Dropout Rates

	No FEs			**School FEs**		**School and Year FEs**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
3-Year Turnover Rate	0.085***	0.006	0.078***	0.015	-0.036**	0.030*	-0.012
	(0.010)	(0.013)	(0.013)	(0.015)	(0.016)	(0.015)	(0.017)
Charter School		8.315***	14.558***				
		(0.926)	(1.452)				
Charter x Turnover			-0.191***		0.106***		0.084***
			(0.026)		(0.027)		(0.027)
Constant	2.168***	2.857***	1.881***	3.399***	3.815***	2.416***	2.849***
	(0.132)	(0.160)	(0.152)	(0.251)	(0.227)	(0.311)	(0.301)
Observations	5772	5772	5772	5772	5772	5772	5772

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. Analysis uses the high school sample. Columns (1) to (3) use robust standard errors while standard errors for columns (4) to (7) are clustered by school.

Table A7: Effects of Turnover on 4-Year High School Graduation Rates

	No FEs			**School FEs**		**School and Year FEs**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
3-Year Turnover Rate	-0.207***	-0.106***	-0.268***	-0.018	-0.001	-0.027	-0.010
	(0.019)	(0.023)	(0.027)	(0.021)	(0.018)	(0.022)	(0.017)
Charter School		-10.602***	-23.150***				
		(1.536)	(2.338)				
Charter x Turnover			0.397***		-0.035		-0.035
			(0.047)		(0.042)		(0.041)
Constant	88.934***	88.088***	90.253***	85.791***	85.651***	84.726***	84.543***
	(0.279)	(0.314)	(0.345)	(0.353)	(0.276)	(0.582)	(0.509)
Observations	6296	6296	6296	6296	6296	6296	6296

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. Analysis uses the high school sample. Columns (1) to (3) use robust standard errors while standard errors for columns (4) to (7) are clustered by school.

Table A8: Effects of Turnover on Average SAT Reading Scores

	No FEs			**School FEs**		**School and Year FEs**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
3-Year Turnover Rate	-1.173***	-0.826***	-1.448***	0.191***	0.256***	-0.020	-0.106*
	(0.068)	(0.082)	(0.154)	(0.050)	(0.077)	(0.039)	(0.058)
Charter School		-34.788***	-76.777***				
		(3.682)	(4.536)				
Charter x Turnover			1.418***		-0.122		0.158**
			(0.168)		(0.101)		(0.079)
Constant	497.986***	495.252***	503.119***	475.845***	475.318***	494.261***	495.074***
	(1.162)	(1.214)	(1.977)	(0.791)	(0.940)	(0.876)	(0.958)
Observations	4213	4213	4213	4213	4213	4213	4213

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2016-2017 through 2022-2023 school years. No data from the 2019-2020 school year, as no standardized tests were administered. Analysis uses the high school sample. Columns (1) to (3) use robust standard errors while standard errors for columns (4) to (7) are clustered by school.

Table A9: Effects of Turnover on Average SAT Math Scores

	No FEs			**School FEs**		**School and Year FEs**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
3-Year Turnover Rate	-1.296***	-0.969***	-1.710***	0.348***	0.424***	0.052	-0.069
	(0.076)	(0.090)	(0.165)	(0.069)	(0.098)	(0.053)	(0.065)
Charter School		-32.714***	-82.776***				
		(4.035)	(5.098)				
Charter x Turnover			1.691***		-0.143		0.221**
			(0.187)		(0.137)		(0.100)
Constant	487.661***	485.089***	494.469***	461.156***	460.542***	483.573***	484.714***
	(1.326)	(1.380)	(2.170)	(1.113)	(1.233)	(1.146)	(1.116)
Observations	4213	4213	4213	4213	4213	4213	4213

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2016-2017 through 2022-2023 school years. No data from the 2019-2020 school year, as no standardized tests were administered. Analysis uses the high school sample. Columns (1) to (3) use robust standard errors while standard errors for columns (4) to (7) are clustered by school.

Table A10: Effects of Non-Imputed Turnover on Student Outcomes

	(1) ELA Proficiency	(2) Math Proficiency	(3) Attendance Rate	(4) Dropout Rate	(5) 4-Year Grad. Rate	(6) Avg. SAT Reading	(7) Avg. SAT Math
3-Year Turnover Rate	-0.060***	-0.048***	0.003	-0.023	0.002	-0.131**	-0.119*
	(0.012)	(0.011)	(0.003)	(0.017)	(0.020)	(0.065)	(0.069)
Charter x Turnover	0.070***	0.093***	0.019**	0.055	-0.001	0.237**	0.393***
	(0.022)	(0.025)	(0.009)	(0.037)	(0.061)	(0.104)	(0.144)
Constant	36.778***	26.928***	94.369***	3.058***	84.391***	496.922***	487.221***
	(0.247)	(0.215)	(0.055)	(0.297)	(0.519)	(1.035)	(1.134)
School Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	28193	28177	33152	5623	6152	4077	4077

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data sourced from the Illinois State Board of Education over the 2014-2015 through 2022-2023 school years. No SAT data prior to the 2016-2017 school year, as the test did not take its current format. No ELA proficiency, math proficiency, or SAT data from the 2019-2020 school year, as no standardized tests were administered. Standard errors are clustered by school.